Aarush Ram Anandh

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EXECUTIVE SUMMARY

Audio ML developer building Hindustani raag classification with Python, PyTorch and STFT features.
Hands-on resampling, normalisation, windowed inference and interactive prediction diagnostics.
Diffusion and physiological-signal research; asthma detection contribution at about 70% sensitivity.

EDUCATION

Ashoka University — B.Sc. (Hons.), Computer Science and Mathematics **Expected 2028**

AUDIO MACHINE LEARNING PROJECT

MEIKURAL: Hindustani Raag Recognition **Ongoing**
Audio Classification / Signal Processing [GitHub]

- Developing a **Python/PyTorch** pipeline for **seven Hindustani raags** using the Saraga Hindustani dataset; implemented resampling, RMS normalisation and overlapping **30-second audio windows**.
- Explored per-window and full-recording tonic alignment against native-frequency **STFT/CNN** representations; trained a native-STFT CNN on **3,471 windows over 30 CUDA epochs**.
- Built interactive clip inference, prediction timelines, and recording-level and window-level confusion-matrix reports.

RELEVANT RESEARCH EXPERIENCE

Constrained Image-Synthesis Lab, Ashoka University **Sep 2025–Present**
Research Fellow

- Improved latent-diffusion colour fidelity to $\Delta E_{00} < 1$, with approximately **4× better colour accuracy** than evaluated open-source baselines; results reported in the preprint below.
- Developing **Controlled Generalised Sliced Wasserstein (CGSW)**, an optimal-transport framework with control-conditioned probes, projections and slice aggregation; paper submitted to **JMLR**.

Koita Centre for Digital Health, Ashoka University **May 2026–Present**
Research Intern (part-time)

- Contributed to a wearable-data pipeline achieving approximately **70% sensitivity (recall)** for **asthma exacerbation detection** in study evaluation.
- Processed **12.1 million HR and 736,000 HRV readings**; built smoothing, residual, change-point and windowed features, with patient-wise evaluation across five seeds.

ADDITIONAL IMPLEMENTATION EXPERIENCE

Shooting Stability Trainer **Sep 2025**
Computer Vision Prototype [GitHub]

- Prototyped webcam-based motion tracing with **OpenCV, Flask-SocketIO and JavaScript**; used HSV thresholding, contour filtering and centroid estimation to track a coloured marker.
- Built a browser interface with segmentation-mask feedback, adjustable HSV thresholds, two-point scale calibration, aim-centre selection and manual motion-trace recording.

Avartan Labs **May–Jul 2026**
Product Engineering / Machine Learning Intern

- Raised rare-class correct detection from **40% to 80%** (**+40 percentage points**) in an **RT-DETR** detection, classification and tracking pipeline for **18 waste-material classes**.
- Investigated focal loss, logit adjustment, class-specific thresholds and targeted fine-tuning for rare classes; evaluated detector failure modes through crop-based refinement.

PUBLICATION

Inference-Time Loss-Guided Colour Preservation in Diffusion Sampling (2026).
Public preprint: arXiv:2601.17259; colour-fidelity research described above.

TECHNICAL SKILLS

Programming: Python, C++, Bash. **Frameworks:** PyTorch, TensorFlow, Diffusers, OpenCV.
Signal / Prototype Work: STFT, RMS normalisation, resampling, CNNs, Flask-SocketIO, JavaScript.